HANDS-ON ACTIVITY FOCUS GUIDE

Observe INIT, PLAY, and STOP

Follow execution through the Driver Station states

25 MIN	8	STATE OBSERVATION	1
ACTIVITY	STUDENT ROLES	FORMAT	SAFETY LEAD

ACTIVITY OUTCOME Students identify where execution is located during INIT, explain why the robot does not drive before PLAY, and show how STOP makes the active condition false.

Materials

- Robot secured on blocks; gamepad and Driver Station connected.
- BasicTwoMotorTeleOp.java displayed with telemetry and waitForStart() visible.
- Printed state-observation sheet from this guide.
- Cards labeled BEFORE INIT, INITIALIZED, WAITING, ACTIVE, and STOPPED.
- POWERING ON safety poster displayed beside the robot area.

Before students enter

- Confirm Status: Initialized is present before waitForStart() in the source code.
- Verify both wheels are clear and define the no-entry boundary.
- Practice locating the Driver Station indicators for INIT, PLAY, STOP, and disabled state.
- Keep joystick input centered before beginning.

SAFETY BOUNDARY Students never cross into the robot test area while the OpMode is enabled. STOP and visually confirm DISABLED before anyone approaches.

Student setup and roles

Assign one primary role to each student. Rotate roles on a second trial if time permits. The safety lead may stop the activity at any time.

STUDENT	ROLE	RESPONSIBILITY
1	Safety lead	Controls the call-and-clear sequence.
2	Driver Station	Presses INIT, PLAY, and STOP when authorized.
3	Code tracker	Points to the instruction where execution is located.
4	Telemetry reader	Reads each displayed telemetry message aloud.
5	Wait gate	Blocks progress until PLAY is pressed.
6	Joystick tester	Moves one stick only when the teacher authorizes it.
7	Recorder	Completes the state-observation table.
8	Stop verifier	Confirms the OpMode ended and the robot is DISABLED.

Ground rules

- Predict before testing; record evidence before proposing a correction.
- Only the Driver Station operator touches INIT, PLAY, or STOP.
- Any student may call STOP for unexpected motion, instability, heat, noise, or an unclear boundary.
- Change only one variable between tests so the evidence remains interpretable.

POWERING ON gate

SAY AND DO ALL SIX STEPS 1. Secure the robot. 2. Confirm DISABLED. 3. Look around. 4. Make sure everyone is clear. 5. Call POWERING ON! and wait. 6. Visually confirm clear, then enable/PLAY.

After every test: press STOP, visually confirm DISABLED, and only then permit anyone to approach the robot.

25-minute teacher walkthrough

1. Predict four states | 4 minutes

TEACHER	*“Place the four state cards in order. Ask students to predict whether motor commands can repeat in each state.”*

STUDENT	*“The recorder writes predictions for before INIT, after INIT, after PLAY, and after STOP.”*

ASK	*“Which event releases waitForStart()?”*

LISTEN FOR PLAY releases the wait; STOP can also cause the wait to return without making the OpMode active.

TEACHING NOTE Make students commit to predictions before touching the Driver Station.

2. Press INIT and locate execution | 6 minutes

TEACHER	*“With the robot disabled and secured, select the OpMode and press INIT. Ask the code tracker to follow runOpMode() until execution pauses.”*

STUDENT	*“The code tracker points through hardware mapping and telemetry, then stops at waitForStart(). The telemetry reader reads Status: Initialized.”*

ASK	*“Does Initialized mean the control loop is running?”*

LISTEN FOR No. Setup reached the waiting point, but PLAY has not released execution into active control.

TEACHING NOTE If initialization fails, stay disabled and return to Lesson 2's mapping checks.

3. Test the waiting gate | 4 minutes

TEACHER	*"Ask the joystick tester to move a stick before PLAY while everyone watches from outside the boundary."*

STUDENT	*"Students confirm that no active-loop motor response occurs. The wait-gate student continues to block the code path."*

ASK	*"Why can the gamepad move without moving the wheel?"*

LISTEN FOR The statements that read the stick and call setPower() are after waitForStart() and are not executing yet.

TEACHING NOTE Do not confuse no motion with proof of complete safety; the robot remains treated as energized equipment.

4. Release the gate with PLAY | 7 minutes

TEACHER	*"Run the complete POWERING ON sequence. After the pause and visual clear check, authorize PLAY and one gentle stick movement."*

STUDENT	*"The wait gate opens. The code tracker moves to opModeIsActive() and the loop. The joystick tester makes one gentle command and returns to center."*

ASK	*"What changed in the program at PLAY?"*

LISTEN FOR waitForStart() returned, the active condition became true, and loop instructions could execute.

TEACHING NOTE Use only enough input to make the state change visible.

5. Press STOP and prove the exit | 4 minutes

TEACHER	*"Direct the Driver Station operator to press STOP. Ask the stop verifier for two pieces of evidence."*

STUDENT	*"The verifier confirms the wheel stops and the Driver Station shows the OpMode is no longer active/robot is disabled. The code tracker leaves the loop."*

ASK	*"What value does opModeIsActive() have now?"*

LISTEN FOR False, so another loop iteration does not begin.

TEACHING NOTE Confirm DISABLED before anyone crosses the boundary.

Program-state observation record

Student/team: ______________________________ Date: _____________ Robot: _________________

STATE	CODE LOCATION	MOTOR LOOP?	OBSERVED EVIDENCE
Before INIT	Not running	No	
After INIT	Setup -> waitForStart()	No	
Stick moved before PLAY	Still waiting	No	
After PLAY	Active check / while loop	Yes	
Stick returned to center	Loop continues	Yes; power may be 0	
After STOP	Loop exited	No	
DISABLED confirmed	OpMode ended	No	Initials: ____

STUDENT EVIDENCE RULE Write what was commanded and what was observed. Do not write a correction until STOP has been pressed and DISABLED has been confirmed.

Safety sign-off

Before live testing, the safety lead confirms: robot secure [ ] DISABLED [ ] boundary clear [ ] POWERING ON called and wait completed [ ]

After testing: STOP pressed [ ] DISABLED visually confirmed [ ] Safety lead initials: __________

COMPLETION CRITERIA The student uses lesson vocabulary, predicts before testing, records evidence, explains one result from the code or math, and follows STOP/DISABLED rules before approaching the robot.

Debrief

ASK	LISTEN FOR
Why can Initialized be displayed before the robot can drive?	The telemetry was sent before waitForStart(); execution is paused until PLAY.
What is the difference between centered sticks and STOP?	Centered sticks request zero power while the loop still runs; STOP ends the OpMode.
What does telemetry.update() do?	It transmits the current telemetry packet to the Driver Station.
What proves it is safe to approach?	The team presses STOP and visually confirms DISABLED; stopped wheels alone are not enough.

Troubleshooting without guessing

OBSERVATION	NEXT CHECK AFTER STOP/DISABLED
No Initialized message	Check whether mapping failed and whether telemetry.update() was reached.
Wheel moves before PLAY	Press STOP immediately; verify the selected code and investigate unexpected hardware behavior.
PLAY pressed but no response	Check active state, gamepad connection, telemetry, mapping, and commands in that order.
Wheel stops but OpMode remains active	Recognize a centered/zero command; use STOP to end the OpMode.

Technical notes for the teacher

- The FTC runtime invokes runOpMode() during initialization for a LinearOpMode.
- waitForStart() blocks until START or until a stop request causes it to return.
- telemetry.addData() prepares an item; telemetry.update() sends the current packet.
- The outer active check prevents loop entry if STOP was requested before PLAY.
- After STOP, opModeIsActive() is false and the runtime disables hardware outputs.

EXIT TICKET Why can Status: Initialized be true while joystick movement still produces no wheel response?